

Practical-9

```
himasree@LAPTOP-41COT5AA:~/OSLab$ cd ~/OSlab
nano filecopy.c
-bash: cd: /home/himasree/OSlab: No such file or directory
himasree@LAPTOP-41COT5AA:~/OSLab$ nano source.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ cat source.txt
This is a test file for the Linux file copy utility.

This program demonstrates low-level file I/O using:
open()
read()
write()
lseek()
close()

It also compares these system calls with:
fopen()
fread()
fwrite()
fclose()Hello World
This is a test file.
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ cat filecopy.c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/time.h>

#define BUFFER_SIZE 4096

double get_time() {
    struct timeval tv;
    gettimeofday(&tv, NULL);
    return tv.tv_sec + tv.tv_usec / 1000000.0;
}

/* Low-level system call copy */
void low_level_copy(const char *source, const char *destination) {
    int src, dest;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    src = open(source, O_RDONLY);

    if (src < 0) {
        perror("open source");
        exit(1);
    }

    dest = open(destination, O_WRONLY | O_CREAT | O_TRUNC, 0644);
```

```

    if (dest < 0) {
        perror("open destination");
        close(src);
        exit(1);
    }

    /* Demonstrate lseek() */
    off_t position = lseek(src, 0, SEEK_SET);

    if (position == (off_t)-1) {
        perror("lseek");
    }

    while ((bytes_read = read(src, buffer, BUFFER_SIZE)) > 0) {
        bytes_written = write(dest, buffer, bytes_read);

        if (bytes_written != bytes_read) {
            perror("write");
            close(src);
            close(dest);
            exit(1);
        }
    }

    close(src);
    close(dest);
}

/* Standard library copy */
void standard_copy(const char *source, const char *destination) {

```

```

    FILE *src, *dest;
    char buffer[BUFFER_SIZE];
    size_t bytes_read;

    src = fopen(source, "rb");

    if (src == NULL) {
        perror("fopen source");
        exit(1);
    }

    dest = fopen(destination, "wb");

    if (dest == NULL) {
        perror("fopen destination");
        fclose(src);
        exit(1);
    }

    while ((bytes_read = fread(buffer, 1, BUFFER_SIZE, src)) > 0) {
        fwrite(buffer, 1, bytes_read, dest);
    }

    fclose(src);
    fclose(dest);
}

int main() {
    double start, end;

```

```

printf("=== FILE COPY PERFORMANCE COMPARISON ===\n\n");

/* Low-level copy */
start = get_time();

low_level_copy("source.txt", "copy_lowlevel.txt");

end = get_time();

printf("Low-level system calls:\n");
printf("open(), read(), write(), lseek(), close()\n");
printf("Time: %.6f seconds\n\n", end - start);

/* Standard library copy */
start = get_time();

standard_copy("source.txt", "copy_standard.txt");

end = get_time();

printf("Standard library functions:\n");
printf("fopen(), fread(), fwrite(), fclose()\n");
printf("Time: %.6f seconds\n\n", end - start);

printf("File copying completed successfully.\n");

return 0;
}
himasree@LAPTOP-41COT5AA:~/OSLab$ |

```

```

himasree@LAPTOP-41COT5AA:~/OSLab$ gcc -Wall -g filecopy.c -o filecopy
himasree@LAPTOP-41COT5AA:~/OSLab$ ./filecopy
=== FILE COPY PERFORMANCE COMPARISON ===

Low-level system calls:
open(), read(), write(), lseek(), close()
Time: 0.024870 seconds

Standard library functions:
fopen(), fread(), fwrite(), fclose()
Time: 0.000128 seconds

File copying completed successfully.
himasree@LAPTOP-41COT5AA:~/OSLab$ |

```

```

himasree@LAPTOP-41COT5AA:~/OSLab$ cat copy_lowlevel.txt
This is a test file for the Linux file copy utility.

This program demonstrates low-level file I/O using:
open()
read()
write()
lseek()
close()

It also compares these system calls with:
fopen()
fread()
fwrite()
fclose()Hello World
This is a test file.
himasree@LAPTOP-41COT5AA:~/OSLab$ |

```

```
THIS IS A TEST FILE.
himasree@LAPTOP-41COT5AA:~/OSLab$ cat copy_standard.txt
This is a test file for the Linux file copy utility.

This program demonstrates low-level file I/O using:
open()
read()
write()
lseek()
close()

It also compares these system calls with:
fopen()
fread()
fwrite()
fclose()Hello World
This is a test file.
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ cmp source.txt copy_lowlevel.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ cmp source.txt copy_lowlevel.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ nano redirect.c
himasree@LAPTOP-41COT5AA:~/OSLab$ nano input.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ ./redirect
Opening input.txt...
himasree@LAPTOP-41COT5AA:~/OSLab$ dup2(output_fd, STDOUT_FILENO);
-bash: syntax error near unexpected token `output_fd,'
himasree@LAPTOP-41COT5AA:~/OSLab$ cat output.txt
=== I/O REDIRECTION DEMO ===
Standard input is now redirected from input.txt.
Standard output is now redirected to output.txt.

Contents of input.txt:
Hello from the input file!

This text will be read through standard input.
The dup2() system call redirects stdin from the keyboard
to this file.

Redirection completed successfully.
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```